

ECON 712: Applied Econometrics

Lecture 4: Regression Fundamentals: Algebraic Properties of OLS

Multiple Linear Regression

Linear Regression using Vector Spaces

Multiple Regression: Matrix Form

- We are going to write the Multiple Regression Equation with k regressors and n observations as:

$$\underset{n \times 1}{\mathbf{y}} = \underset{k \times 1}{\mathbf{X}} \underset{n \times 1}{\boldsymbol{\beta}} + \underset{n \times 1}{\mathbf{u}}$$

where

$$\underset{n \times k}{\mathbf{X}} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_n \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1k} \\ x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

- By Matrix Multiplication:

$$y_i = \mathbf{X}_i \boldsymbol{\beta} + u_i \text{ (ETM Notation)}$$

$$y_i = \sum_{s=1}^{k+1} x_{is} \beta_s + u_i$$

Multiple Regression

- To fully specify the previous model, we need to put structure on the distribution of the vector of error terms.
 - ▶ ETM defines as a fully specified model, a model that allows us to simulate random vector $\mathbf{Y}$ given $\mathbf{X}$
- Let's assume that u 's are *iid* and that $E[u_i | \mathbf{X}_i = \mathbf{x}] = E[u_i] = 0$ for all possible values of $\mathbf{X}_i$ (the model is still underspecified), then

$$E[Y_i | \mathbf{X}_i = \mathbf{x}] = \mathbf{x}\beta$$

- ▶ Conditional Mean Independence of the error term makes the linear regression to be equal to the CEF
- ▶ Conditional Mean Independence of the error term for all possible values of $\mathbf{X}_i$ can only be achieved if the CEF is linear
- ▶ Note that we have not said anything about whether β is causal (as opposed to what we said about ρ)

Multiple Regression: Matrix Form

- The Method-of-Moment Estimator is expressed as:

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X} \hat{\beta}) = \mathbf{0}$$

- By matrix multiplication:

$$\sum_{i=1}^n x_{ik} (y_i - \mathbf{x}_i \hat{\beta}) = 0 \quad k \text{ equations}$$

$$\sum_{i=1}^n (y_i - \mathbf{x}_i \hat{\beta}) = 0 \quad 1 \text{ equation}$$

- Solving for these, we obtain:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- $(\mathbf{X}' \mathbf{X})$ can only be inverted if it is full rank (which requires $\mathbf{X}$ to be full rank– i.e. no perfect multicollinearity)

- The same estimator can be derived by solving:

$$\underset{\hat{\beta}}{\text{Min}} (\mathbf{y} - \mathbf{X} \hat{\beta})^T (\mathbf{y} - \mathbf{X} \hat{\beta})$$

Multiple Linear Regression

Linear Regression using Vector Spaces

Vector Spaces

- An interesting way to understand regression analysis is to study it using vector spaces
- There are three concepts in vector geometry that are used to study OLS:
 1. Scalar product
 2. Vector Addition
 3. Subspaces of Euclidean Spaces

Scalar Product

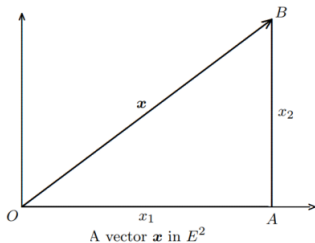
- Scalar product is a single number that determines the relationship between two vectors of the same dimension:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

where both vectors are of size $n \times 1$

- An important property application of the scalar product is the **length** or **norm** of a vector defined as:

$$\|\mathbf{x}\| = \left[\sum_{i=1}^n x_i^2 \right]^{\frac{1}{2}} = \langle \mathbf{x}, \mathbf{x} \rangle^{\frac{1}{2}}$$



Scalar Product

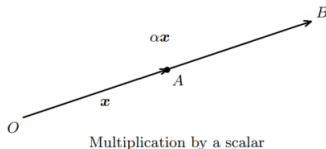
- Scalar product is a single number that determines the relationship between two vectors of the same dimension:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

where both vectors are of size $n \times 1$

- Another important property is that if the angle between two vectors, θ , is zero, then:
 - ▶ This property is based on multiplying a vector by a scalar:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\|$$



Scalar Product

- Scalar product is a single number that determines the relationship between two vectors of the same dimension:

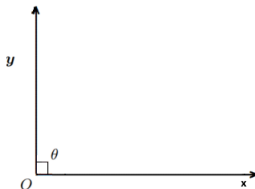
$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

where both vectors are of size $n \times 1$

- Lastly if the angle between two vectors, $\theta = 90^\circ$ (i.e. they are **orthogonal**), then:

► E.g. $\mathbf{x}^T = [1, 0]$, $\mathbf{y}^T = [0, 1]$

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0$$

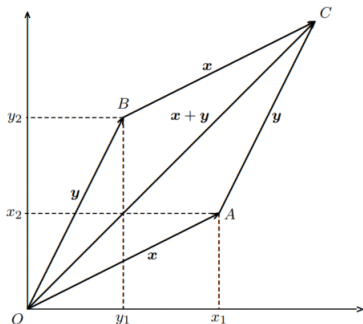


Vector Addition

- Adding two vectors of size n results in:

$$\mathbf{z} = \mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Vector addition in 2 dimensions



- Vector addition can be combined with multiplication by a scalar

Euclidean Spaces and Subspaces

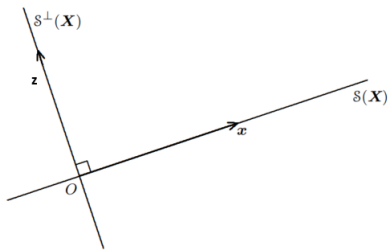
- Vectors of size n can be thought as coordinates of a point in E^n
- The minimum set of vectors that spans a particular space are called the **basis** vectors
 - ▶ We need k linearly independent vectors of size $n \geq k$ to span the k -dimensional space
 - ▶ If $k < n$, we call k a subspace of E^n

Orthogonal Complement

- We call an orthogonal complement, the set of vectors in E^n that are orthogonal to the basis vectors of a k -dimensional subspace.

Example with $n = 2$, $k = 1$

$$\langle \mathbf{x}, \mathbf{z} \rangle = 0$$



The spaces $S(\mathbf{X})$ and $S^\perp(\mathbf{X})$

- These vectors live in the $n - k$ space

Linear Regression

- Linear Regression considers each linearly-independent k column vector of size n from $\mathbf{X}$ as the basis vector that spans the k -dimensional space

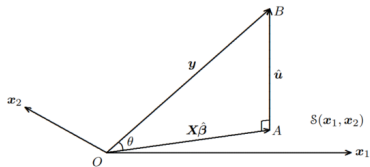
- ▶ We also assume the following two conditions:

$$\mathbf{X}^T \hat{\mathbf{u}} = \mathbf{0} \quad , \quad \mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}$$

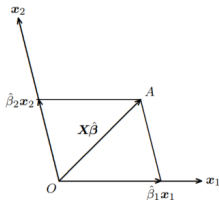
- ▶ This result in linear regression being defined as selecting $\hat{\boldsymbol{\beta}}$ that reaches the perpendicular dropped from $\mathbf{y}$ vector into the plane spanned by $\mathbf{X}$

Linear Regression

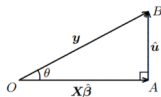
Example with $n = 3$, $k = 2$



(a) y projected on two regressors



(b) The span $S(x_1, x_2)$ of the regressors



(c) The vertical plane through y

Linear regression in three dimensions